

Mini Project 1.2: FDD vs TDD Simulation — Console Output

Output of: python fdd_tdd_sim.py

DL:UL	Scheme	DL thr	UL thr	Util %	Slots DL/UL/G
50:50	FDD	50.0	50.0	100.0	-
	TDD (g=1/20)	45.0	50.0	95.0	9/10/1
	TDD (g=2/20)	40.0	50.0	90.0	8/10/2
	TDD (g=4/20)	30.0	50.0	80.0	6/10/4
70:30	FDD	50.0	30.0	80.0	-
	TDD (g=1/20)	65.0	30.0	95.0	13/6/1
	TDD (g=2/20)	60.0	30.0	90.0	12/6/2
	TDD (g=4/20)	50.0	30.0	80.0	10/6/4
90:10	FDD	50.0	10.0	60.0	-
	TDD (g=1/20)	85.0	10.0	95.0	17/2/1
	TDD (g=2/20)	80.0	10.0	90.0	16/2/2
	TDD (g=4/20)	70.0	10.0	80.0	14/2/4

Crossover (first DL share where TDD utilization exceeds FDD):

guard 1/20 (5% overhead): TDD wins from ~56:44 DL:UL

guard 2/20 (10% overhead): TDD wins from ~61:39 DL:UL

guard 4/20 (20% overhead): TDD wins from ~71:29 DL:UL